

Assignment week 7: From analysis to design

Options for business cases (with descriptions):

1. Library management system (example for inspiration):

- **Description:** This system manages books, students (members), librarians and lending activities in a university library. Students can search for available books, borrow them and return them. Librarians can manage the book stock and handle loan requests.
- **Examples of concepts:** book, student, librarian, borrowing, reserving.
- **Use cases:** Borrow a book, return a book, search the catalogue, reserve a book.
- **Domain objects:** book, student, librarian, loan transaction.

2. Online food ordering system:

- **Description:** This system allows customers to view menus from different restaurants, place orders and track deliveries. Restaurant owners can manage their menus and delivery drivers can update the status of deliveries.

3. Student registration system:

- **Description:** This system allows students to register for courses, view timetables and manage study plans. Teachers can manage course offerings and view registration lists.

4. Parking management system:

- **Description:** This system manages parking spaces for an office complex. Employees can reserve parking spaces, check availability, and manage their reservations. Administrators can allocate parking spaces and ensure efficient use of the parking spaces.

5. Online ticket sales system for events:

- **Description:** On this platform, users can purchase tickets for events such as concerts, sports matches and theatre performances. Event organisers can post events and users can view event details and reserve tickets.

Part 1: Apply the following model: SSD

Apply this week's theory to the business case you chose at the beginning of this academic year. Start with two SSDs of 2 of the use case descriptions from week 2.

Part 2: Now write out OCs:

Then write out 2 OCs of your choice for each SSD, making sure there are postconditions.

Part 3: Individual research task – Sequence diagram

Assignment (for next lesson):

In preparation, each student will individually research **a sequence diagram**

Answer in a short summary:

- What is a Sequence Diagram?
- What components does it contain?
- What is its purpose in software development?

Application:

Draw one **sequence diagram** of your own based on the case your group has chosen.

Use your research as a basis for your diagram.

To be submitted individually:

- A short summary of your findings (±200 words).
- A drawn sequence diagram (digital).